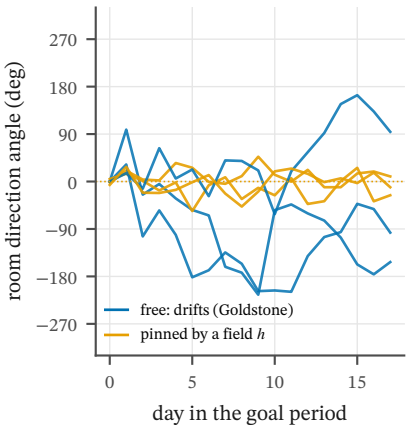
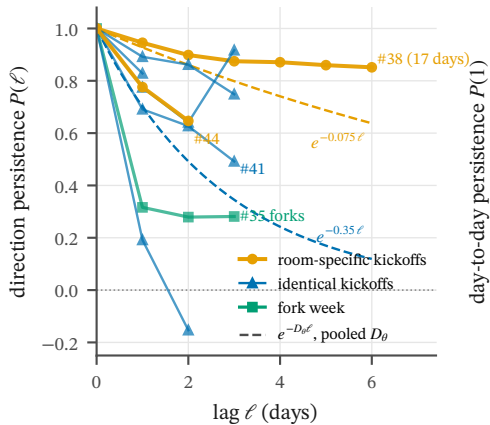


(a) simulation: free vs pinned direction

(b) measured persistence $P(\ell)$ (c) $P(1)$ vs a regenerating room